Design of Enterprise IoT

Tutorial 3 (IoT Network Connectivity)

Read the following use scenario and determine the communication/network technologies for the proposed solution.

IoT Use Case: Intelligent Port System for Fraud Detection

Port operations play a crucial role in global trade, handling vast amounts of cargo daily. However, fraudulent activities, such as cargo theft, pose significant financial and operational risks. According to China Police Daily, between 2013 and 2017, Chinese Public Security Traffic Management Authorities cracked over 4,000 criminal cases related to port logistics (Song et al., 2020). Cargo theft is often considered a "silent crime" because losses frequently go unreported and may only be discovered weeks or even months later.

One common cargo theft tactic involves manipulating a truck's empty weight before entering the port. Fraudsters add materials like liquid tanks, cement, or scrap to artificially increase the truck's weight during the initial weigh-in. After getting the empty truck weighed, these materials are secretly removed, creating a weight discrepancy that allows the truck to load excess cargo undetected. Such fraudulent practices lead to financial losses, regulatory violations, and operational inefficiencies.

Traditional fraud detection methods rely heavily on manual inspections, post-event investigations, and rule-based monitoring systems, which struggle to detect anomalies in real time. These limitations highlight the need for an intelligent port system that enhances security, improves efficiency, and minimizes fraudulent activities. IoT technology offers a promising solution by enabling real-time tracking, automated data collection, and intelligent analytics, allowing ports to proactively detect fraud and optimise operations.

Proposed Solution: An IoT-based intelligent port system leverages advanced IoT technology and data analytics to enhance security and efficiency. Key components include:

- **Smart Weighbridge Systems:** Infrared sensors and weight sensors detect improper truck positioning and compare real-time weight with historical records, ensuring accuracy in cargo measurement.

- **Environmental Context Integration:** Weather sensors provide real-time rainfall data to detect fraudulent weight manipulation involving hidden water tanks.

- **GPS-Based Route Monitoring:** GPS tracking ensures trucks follow designated routes, detecting unauthorised stops or route deviations that may indicate fraud.

- **IoT-Enabled Cargo Surveillance:** Smart cameras, RFID scanners, and environmental sensors verify cargo handling, preventing unauthorised cargo movement.

Design Enterprise IoT

Activity: Start with the list of IoT devices you selected last week for the above Intelligent Port System for the Fraud Detection scenario.

- Determine if the selected devices need to connect with each other and to the network. Justify why connectivity is necessary.
- Determine which communication/network technologies (including protocols) can be used to connect IoT devices to the network.

  [Hint: Consider the following questions when making your decision]

  - What are the range and coverage requirements?
    - E.g., what is the required communication distance?
  - What are the power consumption constraints?
    - E.g., are the devices battery-powered or connected to a power source?
  - What is the required data rate (how much data needs to be transmitted over time)?
  - What are the latency requirements (how fast does data need to be transmitted)?
    - E.g., is real-time or near-real-time communication needed?
  - What are the security and reliability requirements?
  - What are the cost considerations?
    - E.g., what are the budget constraints?

References

Song, R., Huang, L., Cui, W., Óskarsdóttir, M., & Vanthienen, J. (2020). Fraud Detection of Bulk Cargo Theft in Port Using Bayesian Network Models. *Applied Sciences*, *10*(3). https://doi.org/10.3390/app10031056